

Verdant Coil – Phase 1.1: Crawler Movement & Control

Goal: Implement core movement mechanics for the crawler and support basic hazard interaction in a tile-based, turn-driven grid world.

1.1 Breakdown

1.1.1 Create the Core Crawler Scene

- Scene: `Crawler.tscn`
 - Root: `Area2D`
 - Children:
 - `Sprite2D` (placeholder graphic)
 - `CollisionShape2D` (circle or square)
 - Script: `crawler.gd`
 - Define movement vector and state
 - Initial movement can be freeform or arrow-key-based
 - Print debug info to console
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1.1.2 Implement Grid & TileMap

- Add a `TileMap` node to the `ExploreMode.tscn`
 - Use basic placeholder tiles (Flesh, Wall)
 - Set tile size (e.g., 32x32)
 - Optional: create `BaseTile.gd` as abstract base for hazards
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1.1.3 Snap Movement to Grid (Turn-Based)

- Movement becomes discrete:
 - One keypress = one tile move
 - Smooth tile-to-tile transitions
 - Lock out input during movement tween
 - Optional: add `TurnManager.gd` (autoload) for pacing
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1.1.4 Collision Detection

- Prevent movement into:
 - Map edges
 - Wall tiles or other unwalkables
 - Use tile flags or metadata to determine passability
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1.1.5 Tile Interaction Hooks

- Define logic on tile entry:
 - `Acid Pool` : deal damage
 - `Sticky Web` : apply delay or slow
 - `Digest Wall` : block unless upgrade active
 - Use Crawler position + `TileMap.get_cell()` to resolve
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1.1.6 Camera Setup

- Attach `Camera2D` to crawler
 - Set as current camera
 - Enable smoothing if desired
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1.1.7 Debug UI & Feedback

- Add small HUD for:
 - Coordinates
 - Health
 - Active upgrades (stubbed)
 - Optionally: print tile type or hazard effect on entry
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1.1.8 Stub Upgrade Effects

- Stub in basic upgrade toggles (manually activated)
 - `Hardened Skin`
 - `Acid Sac`
 - `Directional Memory`
 - Use simple booleans in `UpgradeController.gd`
 - Tile interaction checks respect upgrade state
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Deliverable: Phase 1.1 MVP

- Controllable crawler that moves on a tile grid
- Hazards and walls interact as expected
- Camera follows player
- UI shows health and toggles
- Code structured for expansion (e.g. TurnManager, Upgrades)